📄 App Dev Project Report

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1. Student Details

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About Me:
I am a student in the IIT Madras BS Degree program with a strong interest in web development and database design. I enjoy building full-stack applications that solve real-world problems.

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2. Project Details

Project Title: Placement Portal

Problem Statement:
To design and develop a web-based Placement Portal that connects students and companies, allowing companies to post job drives and students to apply, while administrators can manage and monitor the process.

Approach:
The application is built using Flask with SQLAlchemy for database management. It supports multiple user roles (Student, Company, Admin). Students can apply to drives, companies can manage applications, and admins can approve or reject company profiles and drives.

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3. AI / LLM Declaration

I used ChatGPT (GPT-5) to assist in:

- **For Some UI designing**

The usage of AI/LLM is around 5%, mainly for guidance and suggestions.

All core logic, integration, UI design, and debugging were implemented manually.

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4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend web framework
SQLAlchemy	ORM for database operations
Jinja2	Dynamic HTML rendering
Bootstrap 5	UI styling and responsiveness
SQLite	Database for storing application
HTML/CSS	Frontend structure and styling

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5. Database Schema / ER Diagram

Tables:

User
Stores all users (student, company, admin)
Fields: id, name, email, password, role, profile_status

Student
Stores student-specific details
Fields: id, student_id (FK), dept, city, skills

Company
Stores company details
Fields: id, company_id (FK), company_name, website, status

Drive
Stores job drives created by companies
Fields: id, company_id (FK), job_title, salary, status

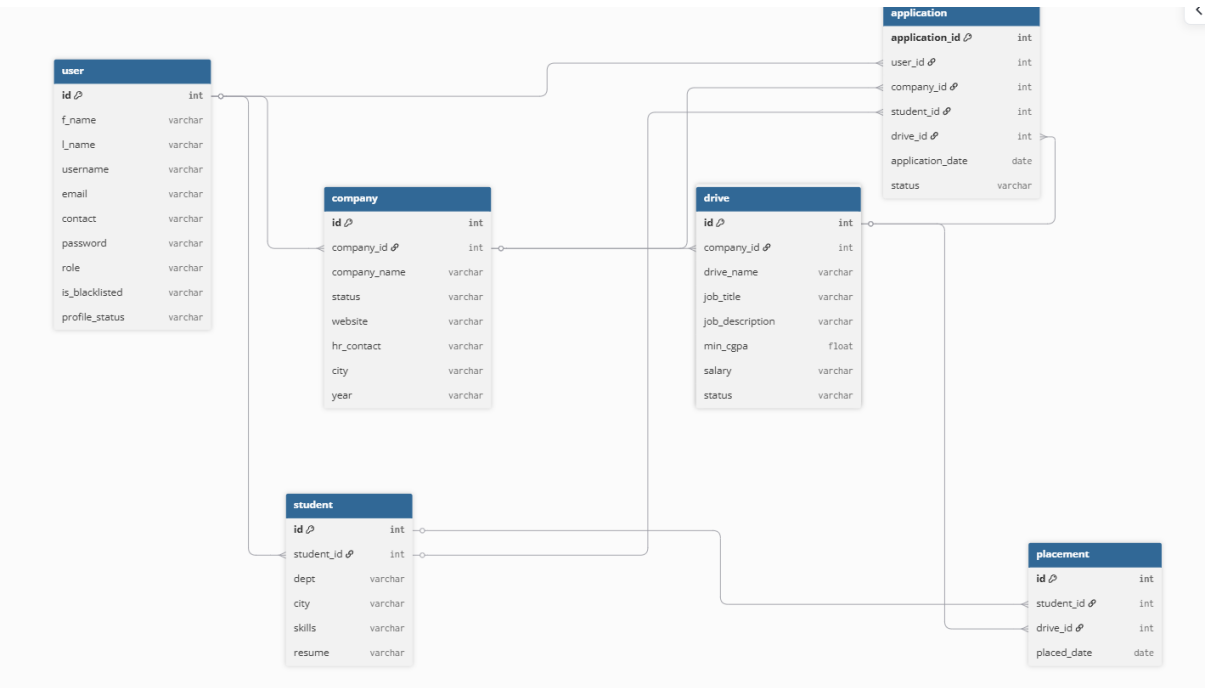
Application
Stores student applications to drives
Fields: application_id, student_id (FK), drive_id (FK), status

Placement
Stores final placement results
Fields: id, student_id (FK), drive_id (FK), placed_date

Relationships:

- **One-to-One → User → Student**
- **One-to-One → User → Company**
- **One-to-Many → Company → Drive**
- **One-to-Many → Drive → Application**
- **One-to-Many → Student → Application**
- **One-to-Many → Student → Placement**

ER Diagram:



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6. Architecture and Features (Optional)

Architecture Overview:

- **app.py → Main Flask app**
- **models.py → Database models**
- **routes/ → All route handlers**
- **templates/ → HTML (Jinja2)**
- **static/ → CSS and assets**

Implemented Features:

✓ User authentication (Student / Company / Admin)
✓ Company profile approval system
✓ Job drive creation and management
✓ Student application system
✓ Application status tracking (Applied, Selected, Rejected)
✓ Admin dashboard for control

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8. Video Presentation

Drive Link:
https://drive.google.com/file/d/16rDktb33LtTq7L53YXkQmNyV1XCq-8xf/view?usp=sharing